

From Data to Decision: Interpreting Neural Network Outputs

Course	MSc in Data Science and AI: Artificial Neural Networks
Name	Shaik Khaja Nayab Rasool
Roll Number	2025em1100102

Objective

Comprehensive analysis & neural network modelling conducted for customer churn prediction in telecommunications sector, to identify customers at risk of churning & to translate gained insights into actionable business decisions.

1. Data Understanding & Preparation

1.1 Dataset Overview

The telecommunications dataset comprises about 7,043 customers with 21 features, categorized into demographics, account details & service usage. The target variable is 'Churn' (Yes/No), indicating whether a customer discontinued service.

1.2 Feature Categories

- Demographics: Gender, SeniorCitizen, Partner, Dependents.
- Account: Tenure (months), Contract type, PaperlessBilling, PaymentMethod.
- Services: PhoneService, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies.
- Charges: MonthlyCharges, TotalCharges.

1.3 Prediction Task

The goal is to build a binary classification model that predicts customer churn by learning patterns from historical customer data, to identify high-risk customers who may discontinue offered service for enabling proactive retention strategies. Each record represents individual customer with attributes such as demographics, account details & service usage. Target variable is Churn, indicates whether customer has discontinued service, which is a binary classification problem.

- 1 (Yes): Customer is likely to churn
- 0 (No): Customer is likely to stay

1.4 Data Cleaning & Preprocessing Highlights

- **Missing Values:** Initial checks revealed no explicit missing values, but 11 empty strings in the TotalCharges column, which corresponded to customers with tenure=0 and were successfully converted to numeric (float64) and imputed as 0.
- **Duplicate Assessment:** Check confirmed no duplicate rows with dataset, ensuring data integrity & uniqueness of each record.
- **Redundancy Handling:** Redundant categorical values such as 'No phone service' & 'No internet service' were standardized to 'No' across relevant service features.
- **Uniqueness:** customerID was confirmed as a unique identifier.
- **No Outliers Removed:** Outliers in numerical features (tenure, MonthlyCharges, TotalCharges) were identified using Z-score & IQR methods but retained, as these were deemed valid customer segments (e.g., long-term high-value customers) rather than data errors, especially given robustness of neural networks to scaled data.

2. Exploratory Data Analysis & Business Insights

Analysis revealed several critical patterns and insights into customer churn behavior. (Visuals: Refer to Notebook sections 2.1 - 2.19 for detailed plots)

- **Churn Distribution:** Dataset shows an imbalance with approximately **26.5% churned** customers against **73.5% retained**, highlighting need for techniques such as SMOTE during modelling.
- **Correlation Heatmap:** Strong positive correlations were observed between tenure & TotalCharges. Both tenure & TotalCharges showed negative correlations with Churn, while MonthlyCharges had moderate positive correlation.

2.1 Key Insights from EDA

- **Early-Stage Customer Vulnerability & High Monthly Charges:** Customers with shorter tenures (especially 0-20 months) and higher monthly charges are significantly more prone to churn. Senior citizens also show higher churn in their initial months.
- **Impact of Contract Type and Payment Method:** Month-to-month contracts & Electronic Check payment methods are strongly associated with higher churn rates. Conversely, longer-term contracts (One year, Two year) and automated payment methods (Bank transfer, Credit card) correlate with higher retention.
- **Influence of Internet Service and Value-Added Services:** Fiber optic internet users exhibit higher churn than DSL users. Customers lacking services like Online Security, Online Backup, Device Protection, or Tech Support are also more likely to churn, indicating their importance for satisfaction and retention.

3. Neural Network Architecture & Development

3.1 Feature Engineering

A new feature, TotalServices, was created by counting subscribed services. Feature selection was performed using Chi-Square method (SelectKBest), identifying top 12 most influential features (e.g., TotalCharges, tenure, MonthlyCharges, Contract types). Less impactful features (e.g., gender, PhoneService) were removed.

3.2 Data Preprocessing for Modeling

- **Encoding:** Binary categorical features were mapped to 0/1, and multi-class categorical features were one-hot encoded.
- **Scaling:** Numerical features (tenure, MonthlyCharges, TotalCharges) were standardized using StandardScaler to bring them to a comparable scale.
- **Class Imbalance:** SMOTE (Synthetic Minority Over-sampling Technique) was applied to the training and validation sets to address the class imbalance, ensuring models learn effectively from both churn and non-churn instances.

3.3 Model Architecture

3 feedforward neural network models **ReLU ELU & SELU** were developed, each with below criteria.

- **Input Layer:** 14 neurons (matching number of selected & processed features).
- **Hidden Layers:** 2 Dense layers (64 neurons, then 32 neurons).
- **Dropout:** Applied after each hidden layer (0.3 and 0.2 rates) to prevent overfitting.
- **Output Layer:** 1 neuron with sigmoid activation function for binary classification.

3.4 Activation Functions

The models differed in their hidden layer activation functions namely ReLU (Model 1), ELU (Model 2), and SELU (Model 3).

3.5 Loss Function & Metrics

All models were compiled using **Adam optimizer**, **binary_crossentropy loss (standard for binary classification)**, evaluated using **accuracy & AUC (Area Under the Receiver Operating Characteristic Curve)** to provide robust performance measure, especially for imbalanced datasets.

3.6 Choice of input features, Loss function, Evaluation metric(s)

3.6.1 Choice of input features

- **Feature Engineering:** A comprehensive process using Chi-Square (SelectKBest) identified features with the strongest statistical relationship to the Churn target. Less impactful features were removed.
- **Domain Knowledge & EDA:** Insights from Exploratory Data Analysis, including correlation heatmaps and bivariate plots, highlighted the predictive power of features like tenure, MonthlyCharges, TotalCharges, Contract type, InternetService, and PaymentMethod.
- **Data Cleaning & Encoding:** All categorical features were properly encoded (binary or one-hot) and numerical features were scaled (StandardScaler), ensuring suitability for the neural network. customerID was dropped as it's not a predictive feature.

3.6.2 Choice of Loss Function

binary_crossentropy was chosen as it is standard and most appropriate loss function for binary classification tasks. It effectively measures the error between the model's predicted probabilities and the actual binary labels (churn or no churn).

3.6.3 Choice of Evaluation Metrics

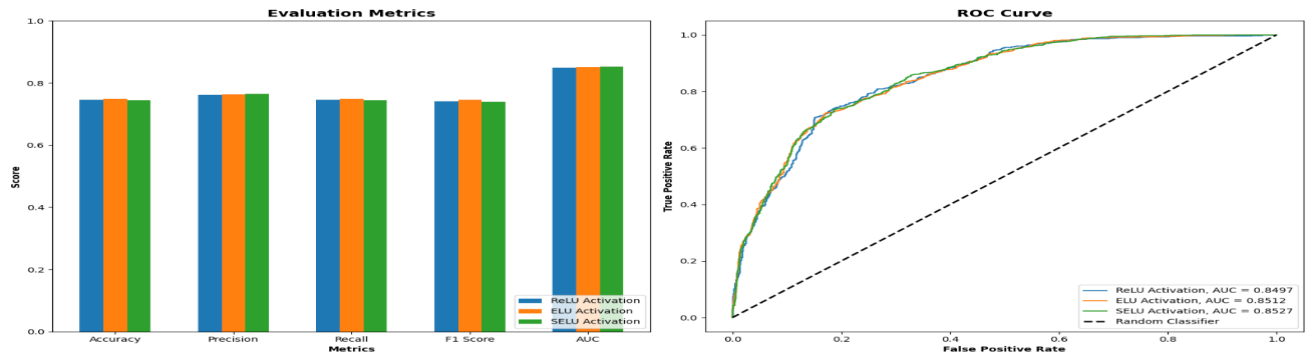
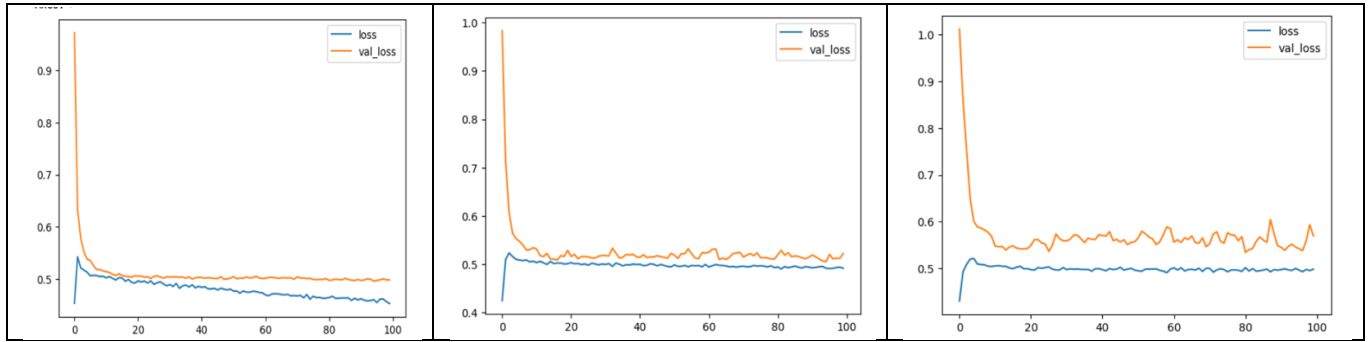
- **accuracy** was included as a general performance indicator to understand the overall correctness of predictions.
- **AUC** (Area Under the Receiver Operating Characteristic Curve) was specifically added because it provides a robust measure of model performance, especially crucial for imbalanced datasets like churn prediction. AUC assesses the model's ability to distinguish between positive and negative classes across various thresholds, considering both true positive and false positive rates.

4. Model Evaluation & Interpretation

(Visuals: Refer to Notebook sections 5.7.1 - 5.7.4 for model training plots, performance comparison, and ROC curves)

4.1 Model Performance Comparison (Validation Set)

	Model	Accuracy	Precision	Recall	F1 Score	AUC
0	ReLU Activation	0.7459	0.7621	0.7459	0.7419	0.8497
1	ELU Activation	0.7498	0.7629	0.7498	0.7466	0.8512
2	SELU Activation	0.7445	0.7647	0.7445	0.7395	0.8527



Model predictions were validated on sample records to ensure consistency between preprocessing pipeline and outputs.

4.2 Chosen Model

modelsig2 (ELU Activation) was selected as best performer due to its marginally superior AUC score (0.8512) and stable training/validation loss curves, indicating good generalization. While all three models performed similarly, ELU demonstrated slight edge in distinguishing between classes.

4.3 What Model Captures Well

- **Overall Churn Prediction:** The model effectively distinguishes churn vs non-churn customers, with AUC ~0.85.
- **Identification of Churners (Recall – Class 1):** High recall (>0.87) indicates strong ability to identify likely churners, supporting proactive retention.
- **Handling Imbalanced Data:** SMOTE enables learning from both churn and non-churn classes, resulting in balanced performance.

4.4 Where Model May Fail (Limitations)

- **Precision for Non-Churners (Class 0):** Lower precision (~0.62–0.64) indicates false positives, where loyal customers may be incorrectly flagged, leading to unnecessary retention efforts.
- **Moderate Overall Accuracy:** ~75% accuracy suggests room for improvement, with some churn drivers not fully captured by the current model or features.
- **Limited Optimization:** Model performance could improve with further hyperparameter tuning (layers, neurons, learning rate) or more advanced architectures.

5. From Insight to Decision

Based on analysis & model performance, two concrete & actionable business decisions are proposed to mitigate customer churn:

5.1 Actionable Decision 1: Proactive Engagement for Early-Stage Customers

1. Insight: Customers with shorter tenures (first 0-20 months), those with higher monthly charges, and senior citizens exhibit significantly higher propensity to churn.

2. Decision: Implement a targeted "New Customer Success Program" with proactive engagement for customers in their initial 20 months, focusing on those with higher monthly bills and senior citizens.

3. Specific Actions: Personalized onboarding calls and follow-ups within the first two weeks, expedited support for new customers, and personalized loyalty incentives after 3-6 months.

4. Assumptions: Early churn is often due to initial dissatisfaction or perceived lack of value. Proactive engagement can improve satisfaction and retention.

5. Risks: Increased operational costs, risk of ineffective personalization or customer fatigue from over-communication.

6. Limitations: Model predicts likelihood, not explicit reasons for dissatisfaction. Program effectiveness depends on core product quality.

7. Conditions for Failure: Unaddressed underlying service quality issues, superior competitor offerings, or retention program costs exceeding customer lifetime value.

5.2 Actionable Decision 2: Strategic Optimization of Payment Methods and Contract Offerings

1. Insight: Month-to-month contracts and Electronic Check payments show higher churn, while longer contracts and automated payments (Bank Transfer, Credit Card) show higher retention.

2. Decision: Incentivize shifts to longer-term contracts and automated payments, while reassessing Electronic Check and month-to-month plans.

3. Specific Actions: Offer discounts or bundles for longer contracts, incentivize automated payments with credits or fee waivers, and analyze Electronic Check user behaviour.

4. Assumptions: Customers are sensitive to financial incentives and convenience. Automated payments reduce passive churn. Longer contracts imply greater commitment.

5. Risks: Initial impact on revenue from incentives, customer resistance to change, or temporary churn delay if core service value is insufficient.

6. Limitations: Correlation does not fully imply causation. Incentive effectiveness depends on design & competitive landscape.

7. Conditions for Failure: Incentives not being compelling enough, economic conditions favouring alternative payment/contract terms, or unaddressed deeper issues with Electronic Check usage.

Conclusion

By combining data-driven insights with strategic business decisions, telecommunications company can move beyond merely predicting churn to actively preventing it. Continuous monitoring of initiatives and iterative refinement based on their impact will be crucial for sustained success in customer retention.